RUTGERS

Semantics IV

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Previously on 201: Set Theory

1. in a set, **order** of items **does not** matter
   - {a, b, c} = {a, c, b} = {b, a, c} = {b, c, a} = {c, a, b} = {c, b, a}

2. **repetitions** of items **do not** matter
   - {a, b, c} = {a, a, a, b, c} = {a, b, b, c, c}

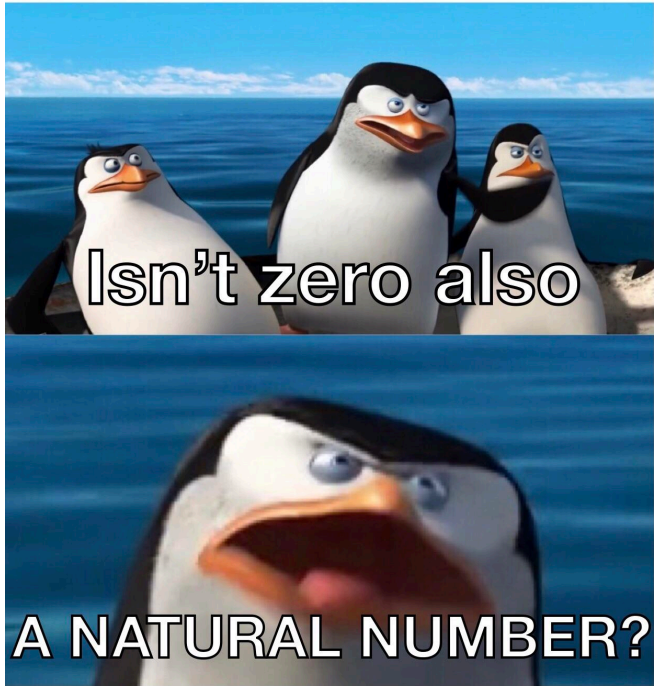
3. an **item** in a set is called the **MEMEBER** of that set
   - and we use symbol $\in$ to describe membership relation

   - a $\in$ {a, b, c}
   - a **is in** / **belongs to** / **is an element of** the set {a, b, c}

$\in$ $\notin$

Teacher: The set of natural numbers is {1, 2, 3, 4, 5...}

Me:



Previously on 201: Set Theory

list notation

the **list notation** specifies a set by listing its members between {curly brackets}

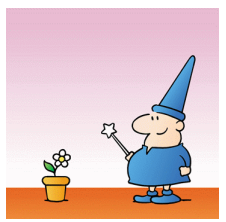
- The set *T* is characters in ***Tom and Jerry***
- *T* = {Tom, Jerry, Tuffy, Spike, Tyke, Butch, Quacker, Cuckoo...}



predicate notation

The **predicate notation** specifies a set by describing member's property

- *S* = { x | x is a sorcerer }
- "*S* equals the set of elements ***x***, such that **every *x*** is a sorcerer"
- normal words: "the set of all sorcerers"
- | stands for "*such that*"



venn diagram notation

Previously on 201: Set Relations

cardinality: the number of elements in a set

- if A = {a, b, c}

  then the **cardinality** of the set A, **|A|** = 3

  also, **|A|** = **| {a, b, c} |**

- **singleton set**: a set with cardinality = 1
- **null set empty set**: a set with cardinality = 0
  - {} = **Ø** "empty set"

subset (⊆)

A is a subset of *B* if **every** element of *A* is **also** an element of *B*

{a, b} ⊆ {a, b, c} {a, b} ⊆ {a, b}

- a ⊆ {a, b, c} is **WRONG**, why?

proper subset (⊂)

A is a proper subset of *B* if *A* is a **subset** of *B* and they are not **identical**

{a, b} ⊂ {a, b, c} {a, b, c} ⊄ {a, b}

identity (=)

- two sets are identical/equal if they have **exactly the same elements**
  - {a, b} = {b, a}

Previously on 201: Set Relations in Languages

Snoopy runs.

- *Snoopy runs.* has the form **Name VERBS**
  - [[Snoopy]] = **element** Snoopy = s

[[Snoppy]] ∈ [[run]]

Cats runs.

- *cats run.* has the form **NOUNs VERB**
  - [[cats]] = **set** of all cats = { x | x is a cat }

[[cat]] ⊆ [[run]]

Compositional Semantics: NPs

- suppose I wanted to use an **NP** like *yellow birds*

What kind of birds do you like?

Yellow birds!

- by using this **NP**, we are not making a **claim**
- we did not say "*birds are yellow*"
- we are just trying to refer to **all elements** that are **simultaneously** in both [[bird]] and [[yellow]]
- to define this in set theory, we need **set operations**

Set Operations

- **set operations** are ways of taking **existing sets** and creating a **new set** out of them
  this is analogous to how **addition** and **subtraction** take **two** numbers and make a **third** number

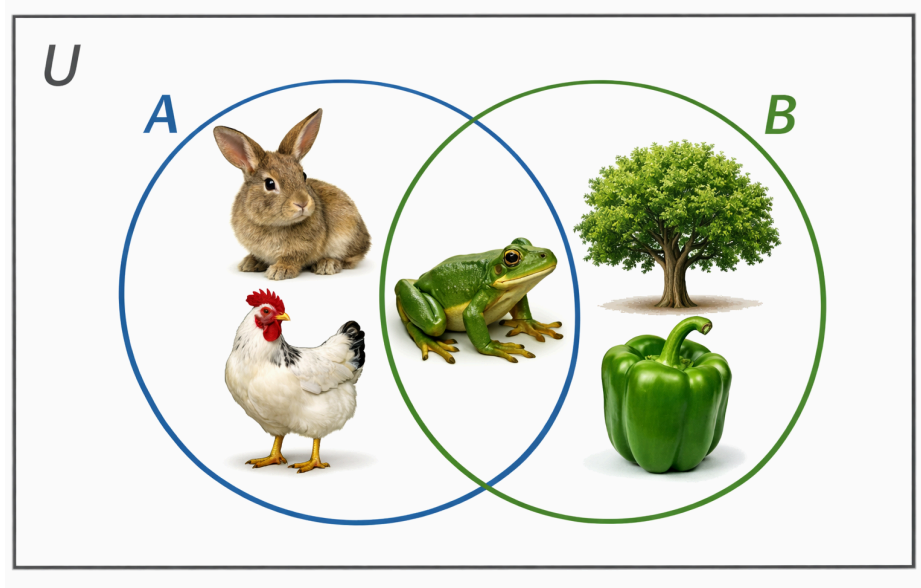
- four set operations: **union (∪)**, **intersection (∩)**, **difference (–)**, **complement (′)**

imagine a **universe U**, in that universe, there are five elements divided into two **sets**

- let A = { x ∈ U | x is an animal } = { rabbit, chicken, frog }

- let B = { x ∈ U | x is green } = { tree, frog, pepper }

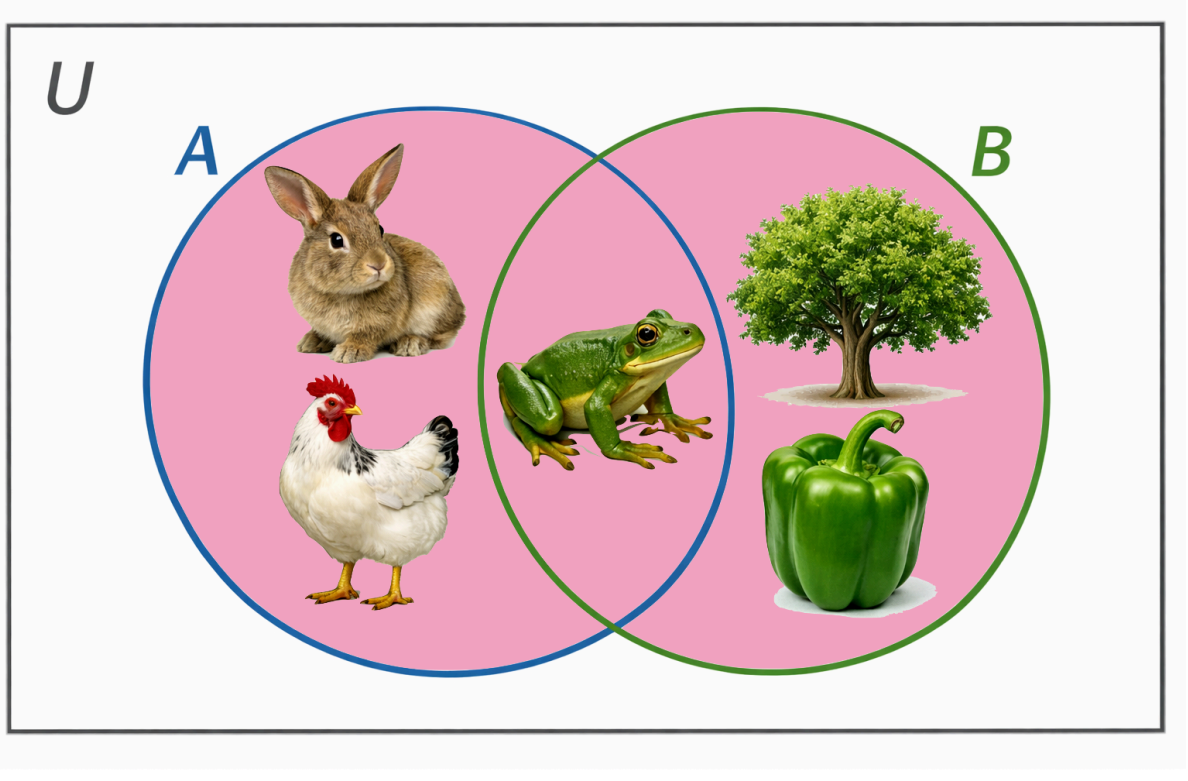
- notice that **frog** is in both A and B



Set Operations: Union

union of A and B produces a new set, which contains all **elements** that **belong to A or B** (or both).

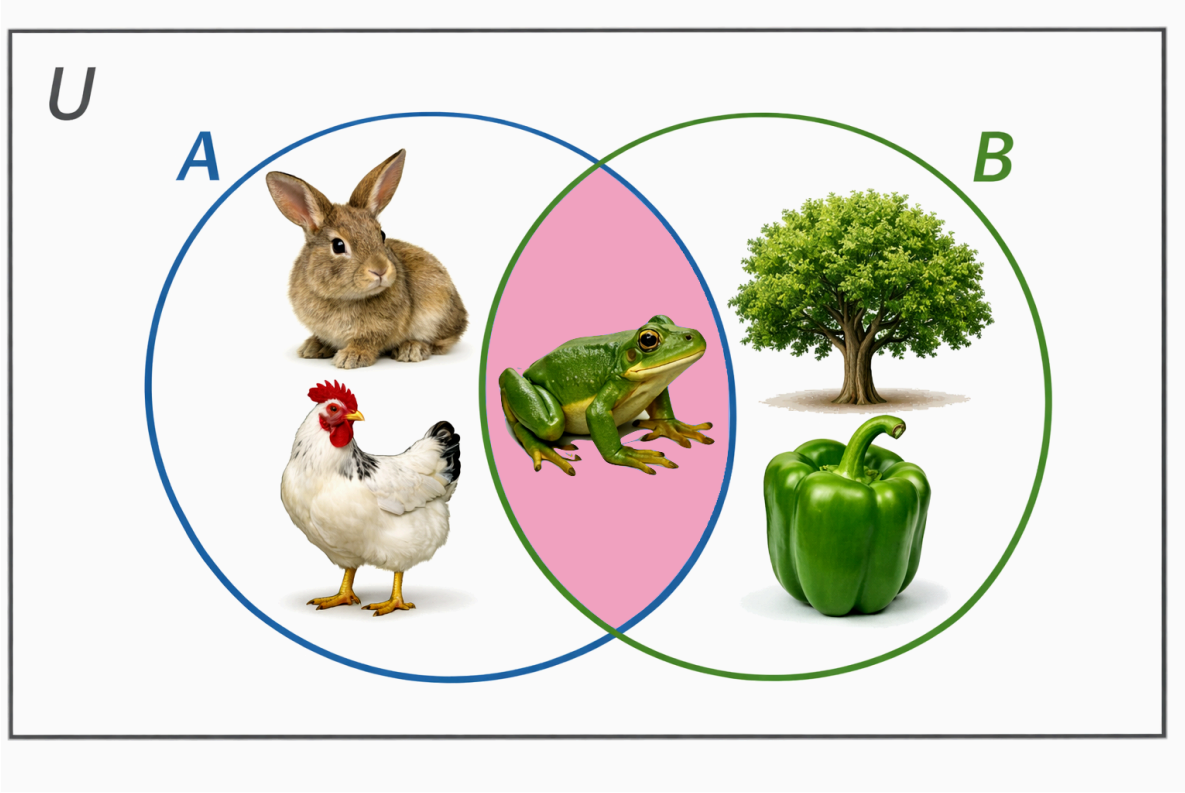
- A ∪ B = { rabbit, chicken, frog, tree, pepper }



Set Operations: Intersection

intersection of A and B produces a new set that contains all elements that **belong to both A and B**

- A ∩ B = { frog }
- **intersection** is the part that **overlaps** between two sets in a Venn diagram

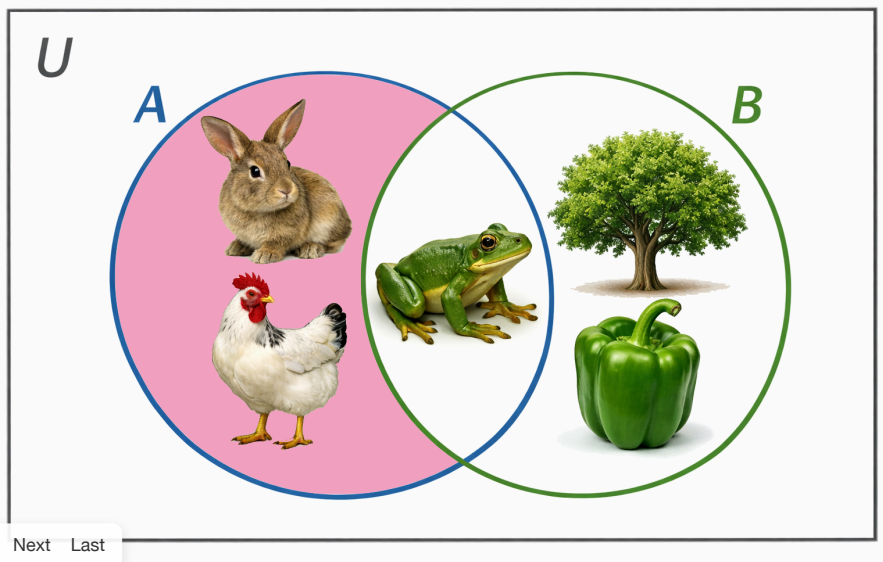


Set Operations: Difference

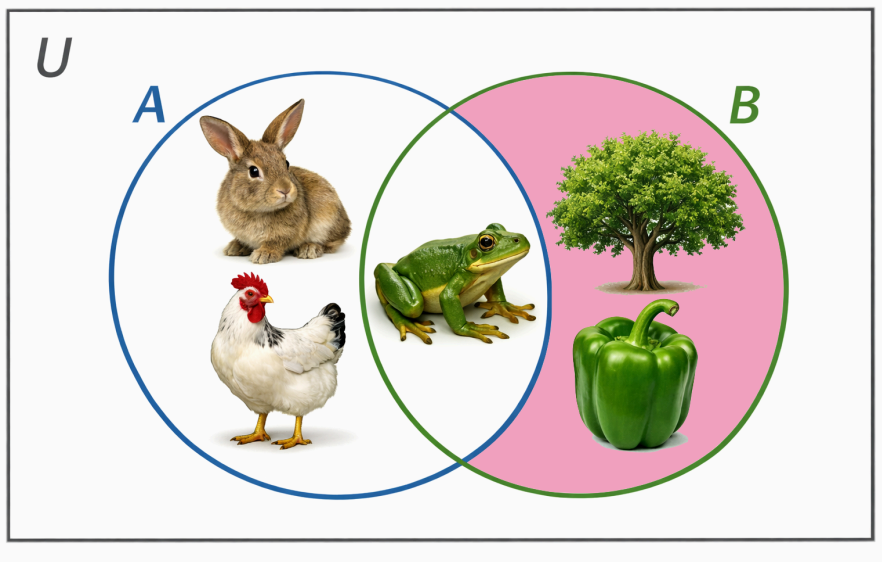
difference (–) of A and B produces a new set, which contains all elements that **belong to A but not B**

- A **–** B = { rabbit, chicken }
- **note** that A **–** B **≠** B **–** A

A – B



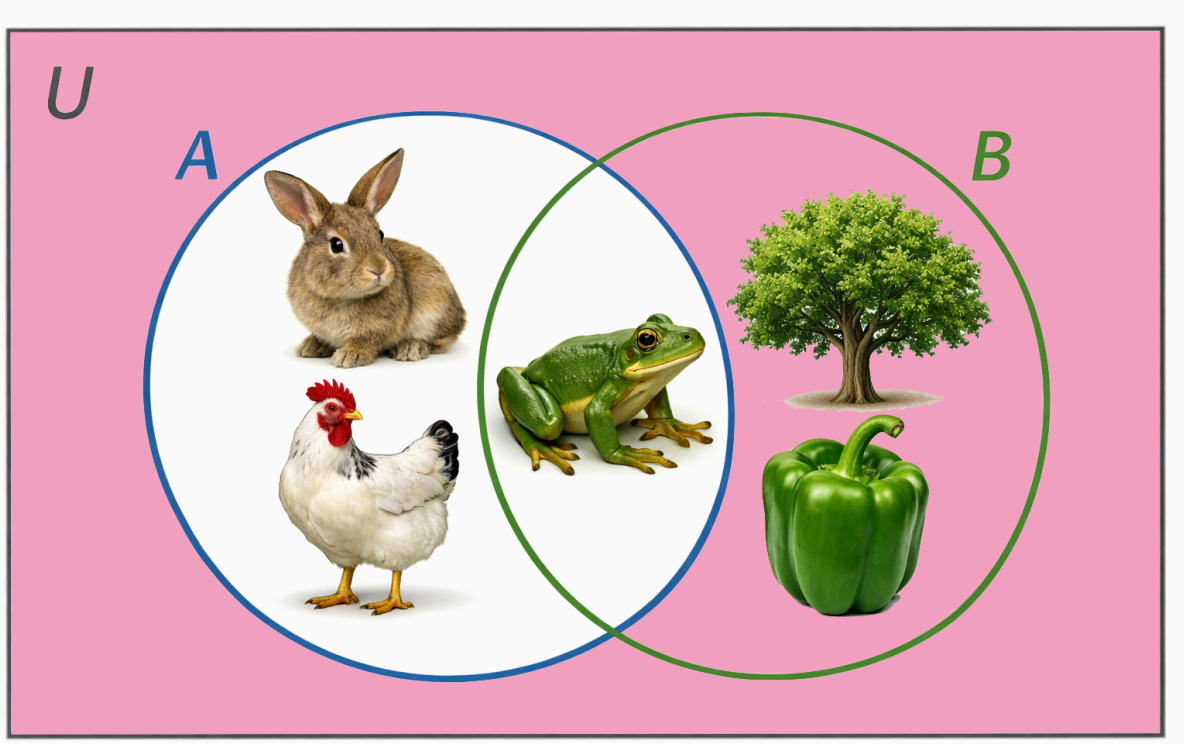
B – A



Set Operations: Complement

complement of A produces a new set, which contains all elements **in the specified universe (U) that are not in A**

- A' = { tree, pepper }
- the symbol of **complement** is tiny, so be careful

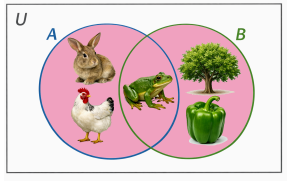


Set Operations: Summary

union (∪)

$A \cup B = \{ x \in U \mid x \in A \text{ or } x \in B \}$

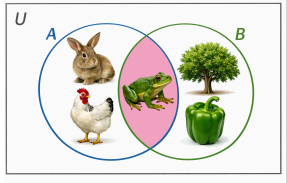
(elements in A or B or *both*)



intersection (∩)

$A \cap B = \{ x \in U \mid x \in A \text{ and } x \in B \}$

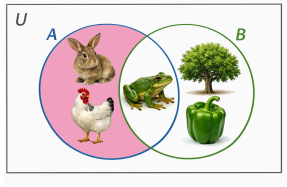
(elements in both A and B)



difference (–)

$A - B = \{ x \in U \mid x \in A \text{ and } x \notin B \}$

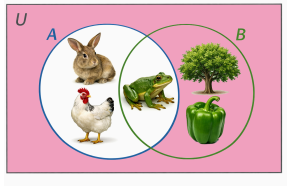
(elements in A but not B)



complement (')

$A' = \{ x \in U \mid x \notin A \}$

(elements in U but not A)



Exercise I: Set Operations

calculate the following formulas, and **name** the set operation(s) involved

1. A ∪ B

2. A – B

3. B ∩ C

4. B ∩ D

5. C′

A = { apple, pear, plum }

B = { orange, pear }

C = { plum, apricot, orange, nectarine }

D = { nectarine, apple }

universe of fruit
U = { apple, pear, plum, orange, apricot, nectarine }

6. (A ∩ B)′

Exercise I: Set Operations

calculate the following formulas, and **name** the set operation(s) involved

1. A ∪ B
   - **union**: {apple, pear, plum, orange}
2. A – B
   - **difference**: {apple, plum}
3. B ∩ C
   - **intersection**: {orange}
4. B ∩ D
   - **intersection**: Ø or {}
5. C′
   - **complement**: {apple, pear}

A = { apple, pear, plum }

B = { orange, pear }

C = { plum, apricot, orange, nectarine }

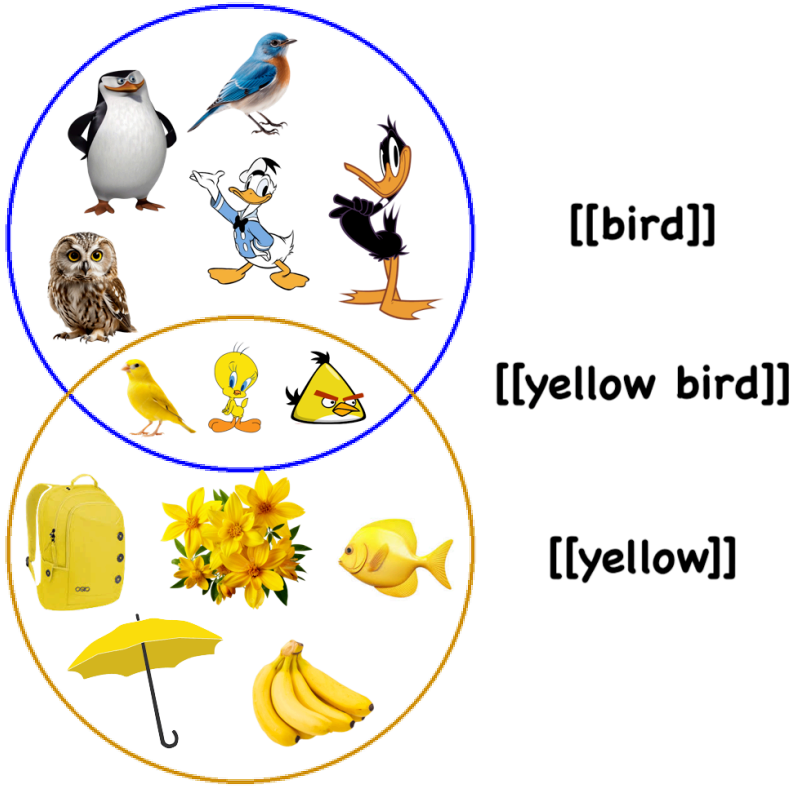
D = { nectarine, apple }

universe of fruit
U = { apple, pear, plum, orange, apricot, nectarine }

6. (A ∩ B)′
   - **intersection**: A ∩ B = {pear}
   - **complement**: (A ∩ B)'= ({pear})' = {apple, plum, orange, apricot, nectarine}

Set Operations in Natural Languages

- returning to our **NP yellow birds**:
  - [[yellow birds]] refers to **all elements** that are in both [[yellow]] and [[bird]]
  - simultaneously in two sets
- what **set operation** do we use to specify elements that are **simultaneously** in two sets?
  - intersection (∩): A ∩ B = all elements that are in both A and B at the same time
  - [[yellow]] ∩ [[bird]] = all elements that are in both [[yellow]] and [[bird]] = [[yellow bird]]
- any **NP** of type ADJ NOUN can be specified in set theory via **intersection**: [[ADJ]] ∩ [[NOUN]]



Set Operations in Natural Languages

- this **NP yellow birds** is further usable in structures we've already seen

Tweety is a bird. [[Tweety]] ∈ [[bird]]

Tweety is a yellow bird. [[Tweety]] ∈ [[yellow]] ∩ [[bird]]

Birds are pretty. [[bird]] ⊆ [[pretty]]

Yellow birds are pretty. [[yellow]] ∩ [[bird]] ⊆ [[pretty]]

- again, this demonstrates the **Principle of Semantic Compositionality**

- more complicated sentences usually need more composing and more set relations



- note that set operators (∪, ∩, −, ′) **take precedence over** set relation symbols (∈, ⊆, =)
- compare this to arithmetic operators (+, −, ×, ÷) and (in)equality symbols (=, <, >) in algebra

Set Theory in Semantics So Far

Union $\cup$

- Brings people together
- Outgoing
- Large and in charge
- Always says yes

Intersection $\cap$

- Great at finding common ground
- Focused
- Sometimes excludes people
- Probably has a LinkedIn account

Superset $\supset$

- Thinks they're better than you
- A bit extra
- Looks at memes all day
- Uses big words to sound smarter

Subset $\subset$

- Shy and insecure
- Kinda unoriginal
- Super cute
- Down to earth

Empty Set $\varnothing$

- Gets jealous easily
- Nihilist
- Super edgy
- Listens to Bon Iver

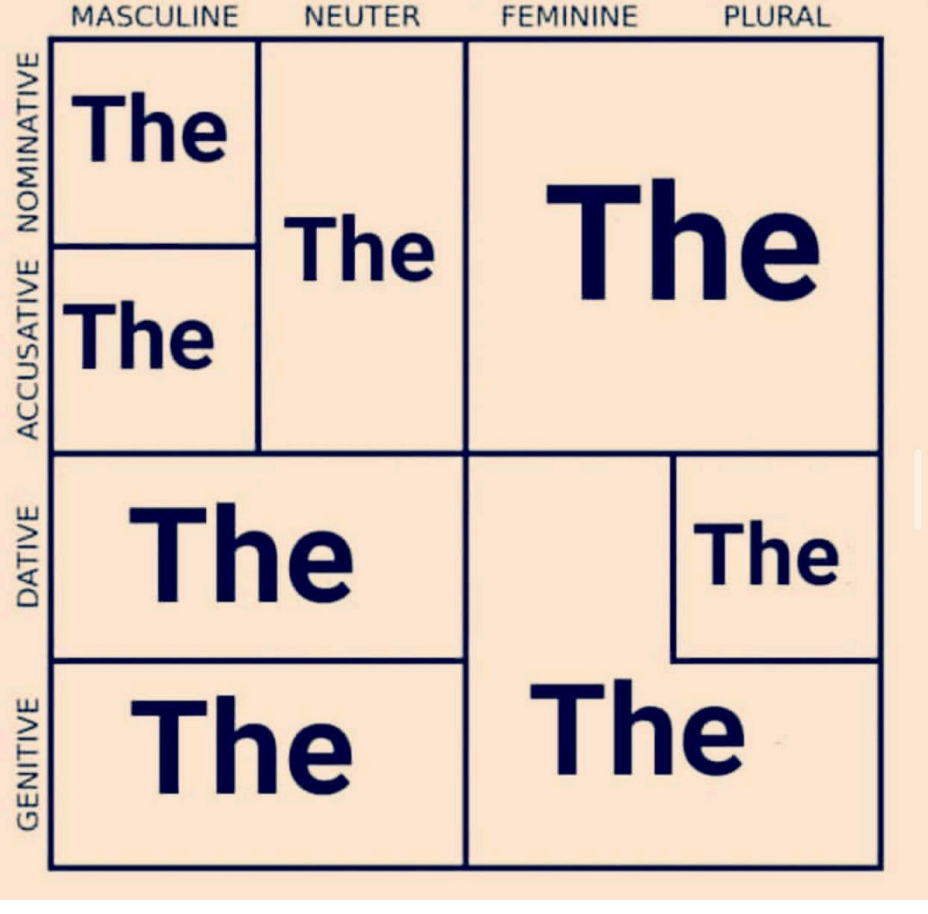
Power Set $\mathcal{P}(X)$

- Know it all
- Leader of the group
- Always wearing a new outfit
- Successful

- we can define words by **reference** – the set of things they refer to
- we can use set theory to precisely define the reference of a **natural language expression**
  - **sets** are **collections** of elements
  - **set relations** can capture the relationship **between words**
    Cats are mammals.(hypo/hypernyms)
    Snoopy is happy.(declarative sentences)
  - **set operations** can capture the reference of **NP**s like ADJ NOUN
- at last, we use**cardinality** to define **quantifiers**

Quantifiers

Semantics of Determiners



- we previously defined **determiner** as a part of speech and it consists of
  - **articles** – *the*, *a*
  - **demonstratives** – *this*, *that*, *these*, *those*
  - **quantifiers** – *every*, *some*, *many*, *most*, *no*
  - **possessive pronouns** – *my*, *your*, *her*, *his*, *their*, *our*
  - some **question words** – *which*, *what*
- determiners serve to **specify** which member of a set we're **referring to** in the world
  - ***my*** *dog* – a **member** of [[dog]] that belongs to me
  - ***this*** *dog* – a **member** of [[dog]] that is near me
  - ***every*** *dog* – the **entire set** of [[dog]]
  - ***the*** *dog* – a **member** of [[dog]] that is relevant to our current conversation
  - ***a*** *dog* – an unspecified **member** of [[dog]]

Semantics of Quantifiers

- **quantifiers** (including numerals) are specific type of **determiners** that describes how many **members** of a set we're talking about

1. **every** cat
2. **some** cats
3. **three** cats
4. **many** cats
5. **most** cats
6. **no** cats



Semantics of Quantifiers

- **quantifiers** (including numerals) are specific type of **determiners** that describes how many **members** of a set we're talking about

1. **every** cat	all
2. **some** cats	more than zero
3. **three** cats	exactly three
4. **many** cats	a large number
5. **most** cats	more than half
6. **no** cats	zero



Semantics of Quantifiers

- **declarative sentences** like NOUNs are NOUNs, NOUNs are ADJ, or NOUN VERBs make some **claim** about the relationship between sets

Cats runs.

⬇

[[cat]] ⊆ [[run]]

Snoopy runs.

⬇

[[Snoppy]] ∈ [[run]]

- sentences with **quantifiers** like QUANT NOUN VERBs also make some **claim** about the relationship between sets – but now it involves some **quantity**

Four cats are running.

  - how many elements are **both** in [[cat]] and [[run]]?
  - the **cardinality** of the intersection of [[cat]] and [[run]] is **four**, in other words, there are **exactly four elements** that are both in [[cat]] and [[run]]

Semantics of Quantifiers

- **declarative sentences** like NOUNs are NOUNs, NOUNs are ADJ, or NOUN VERBs make some **claim** about the relationship between sets

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| [[cat]] ∩ [[run]] | = 4

Semantics of Quantifiers

quantifier	**example**	**set theory expression** (| [[N]] ∩ [[VP]] | = ?)	
numeral N VP	***Two*** *students run*.	| [[N]] ∩ [[VP]] | = n	(n specificed by numeral)
some N VP	***Some*** *students run*.		
every N VP	***Every*** *students runs*.		
many N VP	***Many*** *students run*.		
some N VP	***Some*** *students run*.		
some N VP	***Some*** *students run*.		

Semantics of Quantifiers

quantifier	example	set theory expression (| [[N]] ∩ [[VP]] | = ?)	
numeral N VP	***Two*** *students run*.	| [[N]] ∩ [[VP]] | = n	(n specificed by numeral)
some N VP	***Some*** *students run*.	| [[N]] ∩ [[VP]] | ≠ 0	**or** [[N]] ∩ [[VP]] ≠ ∅
every N VP	***Every*** *students runs*.	| [[N]] ∩ [[VP]] | = | [[N]] |	**or** [[N]] ⊆ [[VP]] = TRUE
many N VP	***Many*** *students run*.	| [[N]] ∩ [[VP]] | > n	(n is some large number)
some N VP	***Some*** *students run*.	| [[N]] ∩ [[VP]] | = ½ | [[N]] |	
some N VP	***Some*** *students run*.	| [[N]] ∩ [[VP]] | = 0	**or** [[N]] ∩ [[VP]] = ∅

Limitations of Set Theory

- so set theory is a useful way to be **precise** about the **semantic meaning** of language
- yet, there are a few things that **cannot** be described in set theory
- an example is **non-intersective** adjectives
  - an adjective like yellow in yellow bird is called **intersective** because [[yellow bird]] can be described as the **intersection** between [[yellow]] and [[bird]]
  - what about *biggest tree* or *matching shirts*?
  - we need more information – for instance to know which tree is biggest, we need to know the size of all other trees we need to compare them
- also, we have not yet differentiated pairs like *Cats run.* and *Cats are running.*
- so **set theory** IS useful, but it **CANNOT** describe all aspects of semantics

Exercise II: Reference Defining

define the reference in terms of sets

1. [[sleep]]

2. [[Arthur]]

3. [[happy]]

4. [[king]]



Exercise II: Reference Defining

define the reference in terms of sets

1. [[sleep]]

[[sleep]] = { x | x sleeps }

2. [[Arthur]]

[[Arthur]] = arthur

3. [[happy]]

[[happy]] = { x | x is happy }

4. [[king]]

[[king]] = { x | x is a king }



Exercise III: Set Relation and Operation

translate sentences into **set theory** expressions

1. Music is beautiful.
2. Beautiful fish swim.
3. Arthur is an ancient figure.
4. Morgana and Merlin are good sorcerers.
5. Dark magic is a horrible thing.
6. Big old dogs are friendly.
7. Every linguist is a mathematician.
8. Most blue cats are elegant.



Exercise IV: Mario Universe

list notation and predicate notation

define the following sets using both **list notation** and **predicate notation**

1. M, the set of characters with mustaches
2. P, the set of characters who are/wear pink
3. C, the set of characters with crowns
4. G, the set of characters who are ghosts

are any of the sets defined above **singletons**?
which one(s)?

U = universe of Mario Kart characters



Exercise IV: Mario Universe

true or **false**? why?

1. Mario ∈ M
2. Peach ∉ C
3. | C | = 3
4. {Daisy, Birdo, Luigi} = {Luigi, Daisy, Birdo}
5. | { x | x has a name that starts with 'W' } | = {Wario, Waluigi}

U = universe of Mario Kart characters



Exercise IV: Mario Universe

use the sets definedbelow, determine if each of these expressions is **true** or **false**

M = { x | x has a mustache} = {Mario, Luigi, Wario, Waluigi}
H = { x | x has a hat} = {Mario, Luigi, Baby Mario, Baby Luigi, Wario, Waluigi}
N = { x | x is a human} = {Mario, Luigi, Baby Mario, Baby Luigi, Peach, Daisy, Wario, Waluigi}
R = { x | x is a princess} = {Peach, Daisy}
F = { x | x is a female human} = {Peach, Daisy}

1.M ⊂ H

2.R ⊆ F

3.H = N

4.N ⊆ M

5.R ⊄ N

6.F ⊈ M

U = universe of Mario Kart characters



A = B: equality/identity
A ⊆ B: subset
A ⊂ B: proper subset

A ≠ B: not equal/identical
A ⊈ B: not a subset
A ⊄ B: not a proper subset

Exercise IV: Mario Universe

given sets on the right, **calculate** the following and **describe** the resulting set in words

01.C ∪ P

02.H ∩ O

03.H – M

04.P ′

05.C ∩ S

06.| F ∪ S |

07.| C ∩ P |

08.O – H

09.(C ∩ F) ∪ M

10.(F ∪ H)′

11.F ∩ S = S ∩ F (true or false?)

12.F – S = S – F (true or false?)

A ∪ *B*: union

A ∩ *B*: intersection

A - *B*: difference

A': complement

U = universe of Mario Kart characters



M = { x | x has a mustache} = {Mario, Luigi, Wario, Waluigi}

H = { x | x has a hat} = {Mario, Luigi, Baby Mario, Baby Luigi, Wario, Waluigi}

C = { x | x has a crown} = {Peach, Daisy, King Boo}

F = { x | x has fangs} = {Bowser, Petey Piranha, King Boo}

P = { x | x is/wears pink} = {Peach, Birdo, Toadette}

S = { x | x has spots} = {Toad, Toadette, Petey Piranha}

O = { x | x's name ends in "o" } = { Mario, Baby Mario, Wario, Birdo, King Boo }

Exercise V: Lion King Universe

write out the **lexical meaning** of these words using **predicate notation**

1. lion
   [[lion]] = L = { x | x is lion }
2. warthog
3. brown
4. meerkat
5. smile
6. ferocious



Exercise V: Lion King Universe

write out the **compositional meaning** of these sentences using **set theory** and decide the **truth value**

1. Timon is a warthog.

   [[Timon]] ∈ [[warthog]]

2. Simba is brown.

3. Meerkats are ferocious.

4. Pumbaa smiles.

5. Lions are not meerkats.



Practice: Lion King Universe

write out the **compositional meaning** of these sentences using **set theory**

1. Simba is a cute lion.
2. Lions are ferocious creatures.
3. Pumbaa is smelly.
4. Adult lions roar.
5. Young lions are adorable animals.
6. Pumbaa is not a sarcastic meerkat.
7. Timon and Pumbaa are friendly.
8. Lions and warthogs are ferocious.



6-8 are **tricky**!

Exercise VI: Stardew Valley Universe

given the definitions to the right, **express** the meaning of the following sentences using **set theory**

1. Abigail is a villager.
2. Mr. Qi does not live in Pelican Town.
3. Every farmer grows crops.
4. Some villagers raise chickens.
5. Only one farmer lives in Pelican Town.
6. Most villagers live in Pelican Town.
7. Marnie and Shane both raise chickens.
8. Neither Robin nor Abigail grow crops.

[[Marnie]] = m
[[Shane]] = s
[[Mr. Qi]] = q
[[Robin]] = r
[[Abigail]] = a

[[villager]] = { x | x is a villager } = V
[[farmer]] = { x | x is a farmer } = F
[[grow crops]] = { x | x grows crops } = G
[[raise chickens]] = { x | x raises chickens } = C
[[live in Pelican Town]] = { x | x lives in Pelican Town } = P

Homework V is due **Mar 29** at **11:59 pm**

reading: **Chapter 8: Pragmatics** of Griffiths & Cummins' (2017) *An Introduction to English Semantics and Pragmatics*



Slides created via the R package **xaringan**.